EE1-10A MATHEMATICS I

Information for Candidates:

Calculators are not permitted in this exam.

1. *Comments about the actual exam performance follow the solution, in italics. Where there are no comments, the questions was done well by most people.*

Express the following complex numbers in the form $x+iy$: [4]

$$\text{(i)}\, z = i^i, \quad \text{(ii)}\, z = \frac{3-i}{2+i}$$

SOLUTION

$$\text{(i)}\, z = i^i = \left(e^{+i\frac{\pi}{2}}\right)^i = e^{-\frac{\pi}{2}}$$

$$\text{(ii)}\, z = \frac{3-i}{2+i}\cdot\frac{2-i}{2-i} = \frac{6-5i+i^2}{4+1} = 1-i$$

Not many correct answers for part (i), almost everyone got (ii). Quite a few people used an interesting approach for (i), involving de Moivre: $i^i = [\cos\pi/2 + i\sin\pi/2]^i = \cos(i\pi/2) + i\sin(\pi/2)$ *and then switching to hyperbolic functions, but few managed to complete it.*

b) Describe and sketch the locus of the complex number which satisfies

$$|z+i| > |z-1|\,. \qquad [\,4\,]$$

SOLUTION

Everything is positive, therefore we can square:

$$|x+iy+i|^2 > |x+iy-1|^2 \;\Rightarrow$$

$$y^2+(y+1)^2 > (x-1)^2+y^2$$

$$2y+1 > -2x+1$$

$$y > -x$$

The locus is $\frac{1}{2}$ of the complex plane $\mathbb{C}$, above the diagonal $y=-x$. A sketch is shown at end.

c) Obtain the limit: [4]

$$\lim_{x\to 0}\ \frac{x^2}{1-\cos x}\,.$$

[Do not use l'Hopital's rule.]

SOLUTION

Writing MacLaurin expansion of $\cos x$

$$\lim_{x\to 0}\ \frac{x^2}{1-(1-\frac{1}{2}x^2+\frac{1}{4!}x^4-\dots)} = \lim_{x\to 0}\ \frac{1}{+\frac{1}{2}-\frac{1}{4!}x^2+\dots} = 2$$

More common approach, the one taken by a large number of correct answers was to multiply top and bottom by $1+\cos x$*, and then apply limit:* $x/\sin x \to 1$.

d) Obtain the limit by any method: [4]

$$\lim_{x\to\pi/3}\frac{\sqrt{3}-\tan x}{1+\cos(3x)}.$$

SOLUTION
The value of the limit is "undetermined", and specifically $\frac{\text{"0"}}{0}$. We can apply l'Hopital's rule.

$$\lim_{x\to\pi/3}\frac{-\sec^2 x}{-3\sin(3x)}=\lim_{x\to\pi/3}\frac{1}{3\sin(3x)\cos^2(x)}=\frac{1}{3\cdot 0\cdot\left(\frac{1}{2}\right)^2}\to\infty$$

The limit does not exist.

Beware of applying l'Hopital without checking that it applies. This works once, because it's an undetermined case of "0/0". But after that, many, many people didn't check, didn't see that it was a case of "1/0", and simply differentiated top/bottom again, arriving at $-8/(9\sqrt{3})$.

e) Differentiate: $y=(\ln x)^x$. [4]

SOLUTION

$$y=(\ln x)^x\to$$
$$\ln y=x\ln(\ln x)$$
$$\frac{1}{y}\frac{dy}{dx}=\ln(\ln x)+x\frac{1}{\ln x}\cdot\frac{1}{x}$$
$$\frac{dy}{dx}=y\left[\ln(\ln x)+\frac{1}{\ln x}\right]=\ln(\ln x)(\ln x)^x+(\ln x)^{x-1}$$

f) Find $\dfrac{dy}{dx}$ when $xy+\ln(xy)=2$. [4]

SOLUTION

$$\frac{d}{dx}(xy)+\frac{d}{dx}(\ln(xy))=0$$
$$1\cdot y+x\frac{dy}{dx}+\frac{1}{xy}\cdot\left[y+x\frac{dy}{dx}\right]=0$$
$$\left(x+\frac{1}{y}\right)\frac{dy}{dx}=-\left(y+\frac{1}{x}\right)$$
$$\frac{dy}{dx}=-\frac{\left(y+\frac{1}{x}\right)}{\left(x+\frac{1}{y}\right)}=-\frac{y}{x}$$

Few people saw the final simplification in the last step. Many also differentiated the l.h.s correctly in the first step, but left a term of 2 on the r.h.s.

g) Integrate by substitution: $\int_0^1 \sqrt{1-x^2}\,dx$. [4]

SOLUTION

Using the substitution $x = \sin u$ we have $\begin{cases} x=0 & \to & u=0 \\ x=1 & \to & u=\frac{\pi}{2} \\ dx = \cos u\,du \end{cases}$

and the integral becomes:

$$\int_0^1 \sqrt{1-x^2}\,dx = \int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2 u}\,\cos u\,du = \int_0^{\frac{\pi}{2}} \cos^2 u\,du$$

$$= \int_0^{\frac{\pi}{2}} [\cos(2u)+1]\,du = \frac{1}{2}\left[\frac{\sin(2u)}{2}+u\right]_0^{\frac{\pi}{2}} = \frac{\pi}{4}$$

A number of people tried a hyperbolic substitution.

h) Integrate: $\int_1^e \frac{(\ln x)^2}{x}\,dx$. [4]

SOLUTION

Using the substitution $u = \ln x$ we have $\begin{cases} x=1 & \to & u=0 \\ x=e & \to & u=1 \\ du = \frac{1}{x}\,dx \end{cases}$

and the integral becomes:

$$\int_1^e \frac{(\ln x)^2}{x}\,dx = \int_0^1 u^2\,du = \frac{u^3}{3}\Big]_0^1 = \frac{1}{3}$$

Difficult if attempting integration by parts here, but a number of those that tried this succeeded. Letting $u = \ln x$ and $dv = (\ln x)/x\,dx$ you obtain

$$I = \int \frac{(\ln x)^2}{x}\,dx = \frac{1}{2}(\ln x)^3 - \frac{1}{2}\int \frac{(\ln x)^2}{x}\,dx = \frac{1}{2}(\ln x)^3 - \frac{1}{2}I$$

and then apply limits.

i) Do the below series converge or diverge? Briefly justify your answer.

$$\text{(I)}\ \sum_{n=1}^{\infty} \frac{n}{3^n}, \qquad \text{(II)}\ \sum_{n=1}^{\infty} \frac{n!}{3^n}, \qquad [\ 4\ ]$$

SOLUTION

We can use the ratio test to study the convergence of both series.

$$\text{(I)}\ \left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{n+1}{3^{n+1}}\cdot\frac{3^n}{n}\right| = \frac{1}{3}\left|\frac{n+1}{n}\right| \underset{n\to\infty}{\longrightarrow} \frac{1}{3}$$

$$\text{(II)}\ \left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)!}{3^{n+1}}\cdot\frac{3^n}{n!}\right| = \left|\frac{(n+1)}{3}\right| \underset{n\to\infty}{\longrightarrow} \infty$$

As $n \to \infty$ the first series converges, while the second diverges.

Many arguments about convergence/divergence without calculations, but often argued a bit too vaguely, such as "terms become smaller" in (I). This is not enough! The terms of the harmonic series also become smaller, but it diverges.

j) Define the odd function $f(t) = \begin{cases} -1 & (-\pi < t \le 0) \\ 1 & (0 < t \le \pi) \end{cases}$ with $f(t+2\pi) = f(t)$ for all t.

Show that the Fourier Series for $f(t)$ is $\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n-1}\sin[(2n-1)t]$.

[4]

SOLUTION

Since $f(t)$ is an odd function, there are no cosine contributions to the Fourier series ($a_n = 0$).

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt = \frac{2}{\pi}\int_{-\pi}^{\pi} 1\cdot sin(nt)\,dt$$

$$= \frac{2}{\pi}\left[-\frac{\cos(nt)}{n}\right]_0^{\pi} = -\frac{2}{\pi n}[(-1)^n - 1] = \begin{cases} \frac{4}{n\pi}, & \text{if } n \text{ is odd} \\ 0, & \text{if } n \text{ is even.} \end{cases}$$

$$f(t) = \frac{4}{\pi}\sum_{\substack{n=1 \\ (n\ odd)}}^{\infty}\frac{1}{n}\sin(nt) = \frac{4}{\pi}\sum_{m=1}^{\infty}\frac{1}{2m-1}\sin[(2m-1)t].$$

Strangely, though many people recognized that the odd function has $a_n = 0$, they still proceeded to calculate a_0. Many, many integrated from $-\pi$ to π. Many didn't quite see how to handle $(-1)^n - 1$, and many errors resulted from trying to force their result into the desired answer which was given.

2. a) Find the value of q for which the limit

$$\lim_{x\to\infty} x^q(\sqrt{x-1}-\sqrt{x})$$

exists and is non-zero. Find the limit. [4]

SOLUTION

$$x^q(\sqrt{x-1}-\sqrt{x}) = x^q\cdot\sqrt{x}(\sqrt{1-\frac{1}{x}}-1)$$

Since $\left|\frac{1}{x}\right| < 1$ as $x\to\infty$ we can use the binomial expansion.

$$= x^{q+\frac{1}{2}}\left[\left(1-\frac{1}{2}\left(\frac{1}{x}\right)+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2}\left(\frac{1}{x}\right)^2+\ldots\right)-1\right]$$

$$= x^{q+\frac{1}{2}}\left[-\frac{1}{2x}-\frac{1}{8x^2}\cdots\right]$$

To have a finite and non zero limit we give $q=\frac{1}{2}$,

$$= -\frac{1}{2}-\frac{1}{8x}+\text{ terms in }\frac{1}{x^3}\,,\,\frac{1}{x^4}$$

and the limit is:

$$\lim_{x\to\infty} x^q(\sqrt{x-1}-\sqrt{x}) = -\frac{1}{2}$$

Many people multiplied top/bottom by $\sqrt{x-1}+\sqrt{x}$. Many found the presence of q difficult to handle.

b) Evaluate the integral $\displaystyle\int \frac{\cosh^{-1}(x)}{\sqrt{x^2-1}}\,dx$. [4]

SOLUTION
We know (formula sheet) that $\frac{d}{dx}\cosh^{-1}x = \frac{1}{\sqrt{x^2-1}}$.
Let $u=\cosh^{-1}x$, $du=\frac{1}{\sqrt{x^2-1}}$ and the integral becomes:

$$\int \frac{\cosh^{-1}(x)}{\sqrt{x^2-1}}\,dx = \int u\,du = \frac{u^2}{2}+c = \frac{1}{2}\left[\cosh^{-1}x\right]^2+c$$

Lots of confusion about this one, with attempts involving $\sinh x$ *or its inverse, often leading to a dead-end. A number took the difficult approach of integrating by parts, leading to*

$$I = \int \frac{\cosh^{-1}(x)}{\sqrt{x^2-1}}\,dx = \left(\cosh^{-1}x\right)^2 - I$$

which works as well.

c) Let $I_n = \displaystyle\int_0^{\pi} \cos^n(x)\,dx\,,\quad n=0,1,2,\ldots.$

i) Integrate by parts to show that $I_n = \dfrac{n-1}{n} I_{n-2}$, for $n \geq 2$. [6]

ii) Using the formula from (i), or otherwise, obtain I_6. [2]

SOLUTION

i)

$$I_n = \int_0^\pi \cos^n x \, dx = \int_0^\pi \cos x (\cos^{n-1} x) x \, dx$$

$$= \sin x \, \cos^{n-1} x \Big]_0^\pi + (n-1) \int_0^\pi \cos^{n-2} x \, \sin^2 x \, dx$$

$$= (n-1) \int_0^\pi \cos^{n-2} x \, (1 - \cos^2 x) \, dx = (n-1) \left[I_{n-2} - I_n \right]$$

$$\Rightarrow I_n \left[1 + n - 1 \right] = (n-1) I_{n-2}$$

$$\Rightarrow I_n = \frac{n-1}{n} I_{n-2}$$

ii)

$$I_0 = \int_0^\pi \cos^0 x \, dx = \int_0^\pi 1 \cdot dx = \pi$$

$$I_2 = \frac{2-1}{2} I_0 = \frac{\pi}{2}$$

$$I_4 = \frac{4-1}{4} I_2 = \frac{3\pi}{8}$$

$$I_6 = \frac{6-1}{6} I_4 = \frac{5\pi}{16}$$

Many, many people tried IBP with $u = \cos^n x$ and $dv = dx$ which led to hopeless entanglement. A few took the (more difficult) approach of writing $\cos^n x = (1 - \sin^2 x) \cos^{n-2} x$ and then considering two integrals, $I_{n-2} - \int \sin^2 x \, \cos^{n-2} x dx$ where the second integral still needed IBP, with $u = \sin x, dv = \sin x \, \cos^{n-2} x dx$, leading eventually to $I_n = I_{n-2} - \frac{1}{n-1} I_n$. In (ii) many people who didn't get (i) then proceeded not to use the reduction formula, even though it was there, and attempted to integrate $\cos^6 x$.

d) Differentiate y to obtain the derivative $\dfrac{dy}{dx}$ for the function

$$y = \ln \left(x + \sqrt{x^2 + 1} \right)$$

and simplify as much as possible. [4]

SOLUTION

$$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2+1}} \left[1 + \frac{1}{2} (x^2+1)^{-\frac{1}{2}} \cdot 2x \right] = \frac{1}{x + \sqrt{x^2+1}} \left[1 + \frac{x}{\sqrt{x^2+1}} \right]$$

$$= \frac{1}{x + \sqrt{x^2+1}} \left[\frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1}} \right] = \frac{1}{\sqrt{x^2+1}}$$

Many people had trouble simplifying to the last step. Odd, considering the integral is on the formula sheet.

3. a) Given the function $f(x) = \dfrac{1}{1+\sin x}$

i) Find $\int f(x)\,dx$ [4]

ii) Using (i) or otherwise, find the area between $f(x)$ and the $x-$axis from $x = 0$ to $x = \pi$, and sketch $f(x)$ on the interval $[0, \pi]$. [4]

SOLUTION

i) Using the substitution $t = \tan\left(\frac{x}{2}\right)$, $dx = \frac{2\,dt}{1+t^2}$ and $\sin x = \frac{2t}{1+t^2}$. The integral becomes:

$$\int \frac{1}{1+\sin x}dx = \int \frac{1}{1+\frac{2t}{1+t^2}} \cdot \frac{2\,dt}{1+t^2} = \int \frac{2\,dt}{1+t^2+2t}$$

$$= \int \frac{2\,dt}{(1+t)^2} = -\frac{2}{1+t} + c = -\frac{2}{1+\tan\left(\frac{x}{2}\right)} + c$$

A non-trivial number of people multiplied top and bottom by $1-\sin x$ giving $\int \tan x - \sec x\,dx$. This worked in some cases, for those that were able to integrate these functions, but many gave up at that point.

ii) Finding the area requires the improper integral

$$\lim_{k\to\pi} \int_0^k f(x)\,dx = \lim_{k\to\pi} \left[-\frac{2}{1+\tan\left(\frac{x}{2}\right)} + \frac{2}{1+\tan 0} \right]$$

As $k \to \pi$, $\tan\left(\frac{k}{0}\right) \to \infty$, so $\frac{2}{1+\tan\left(\frac{x}{2}\right)} \to 0$ leaving

$$Area = \frac{2}{1+0} = 2.$$

The sketch can be found in figure at the end.

Most people ignored the infinite discontinuity in the interval of integration. Many people didn't bother with the sketch, others sketched $\int f(x)\,dx$ instead of $f(x)$!

b) Use De Moivre's Theorem to show that

$$\sin 5\theta = A\sin\theta + B\sin^3\theta + C\sin^5\theta\,,$$

giving the values of A, B, and C. [4]

SOLUTION

$$\cos(5\theta) + i\sin(5\theta) = (\cos\theta + i\sin\theta)^5$$
$$= \cos^5\theta + 5\cos^4\theta\cdot(i\sin\theta) - 10\cos^3\cdot(i\sin\theta)^2 +$$
$$+10\cos^2\theta\cdot(i\sin\theta)^3 + 5\cos\theta\cdot(i\sin\theta)^4 + (i\sin\theta)^5$$
$$= \text{Real part } + i(5\sin\theta\cos^4\theta - 10\sin^3\theta\cos^2\theta + \sin^5\theta).$$

Hence

$$\sin(5\theta) = \sin^5\theta + 5\sin\theta\cdot(1-\sin^2\theta)^2 - 10\sin^3\theta\cdot(1-\sin^2\theta)$$

$$= \sin^5 + 5\sin\theta(1 - 2\sin^2\theta + \sin^4\theta) - 10\sin^3\theta + 10\sin^5\theta$$
$$= 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$$

and $A = 5$, $B = 20$ and $C = 16$.

A large number of people multiplied out $(\cos\theta + i\sin\theta)^2$*, then multiplied the result by* $\cos\theta + i\sin\theta$*...and again, twice. In one or two cases, no errors resulted from this. A worryingly large minority gave* $A = 5\cos^4\theta, B = -10\cos^2\theta, C = 1$.

c) i) Show that $\ln(1+x)$ has Maclaurin series

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + R_4$$

and find the Lagrange remainder R_4 [5]

ii) Show that the radius of convergence for the Maclaurin series in (i) is 1. [3]

SOLUTION

i)

$$f(x) = \ln(1+x) \quad \longrightarrow \quad f(0) = 0$$
$$f'(x) = \frac{1}{1+x} \quad \longrightarrow \quad f'(0) = 1$$
$$f''(x) = -\frac{1}{(1+x)^2} \quad \longrightarrow \quad f''(0) = -1$$
$$f'''(x) = +\frac{2}{(1+x)^3} \quad \longrightarrow \quad f'''(0) = 2!$$
$$f^{(4)}(x) = -\frac{2\cdot 3}{(1+x)^4} \quad \longrightarrow \quad f^{(4)}(0) = 3!$$
$$f^{(5)}(x) = +\frac{2\cdot 3\cdot 4}{(1+x)^5}$$

Maclaurin series is

$$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}}{4!}x^4 + \ldots$$
$$= x - \frac{1}{2}x^2 + \frac{2!}{3!}x^3 - \frac{3!}{4!}x^4 + R_4 = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + R_4$$

and $R_4 = \frac{f^{(5)}(c)}{5!}x^5 = \frac{1}{5(1+c)^5}x^5$, with $0 < |c| < |x|$.

ii) For convergence $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|-\frac{\frac{1}{n+1}x^{n+1}}{\frac{1}{n}n^n}\right| = \left|\frac{n}{n+1}\right||x| < 1 \text{ as } n \to \infty.$$

And since $\lim_{n\to\infty}\frac{n}{n+1} = 1 \Rightarrow |x| < 1$.

A lot of good answers, but the remainder needed to be in the simplified form, many just left it as $f^{(5)}(c)x^5/5!$*.The ratio test needed to be done in full detail, given that the answer is known.*

Figure 4.1 text

4. a) The function $f(x)$ is periodic, with period $T = 2$, and is an even function of x. In the interval $0 < x \leq 1$ the function has the value

$$f(x) = x - 1\,, \qquad 0 \leq x < 1.$$

i) Sketch the graph of $f(x)$ for $-3 < x < 3$. [2]

ii) Show that the Fourier Series for $f(x)$ is

$$f(x) = -\frac{1}{2} - \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2m-1)^2}\cos[(2m-1)\pi x]\,.$$

[6]

iii) Using the results of (ii) and Parseval's theorem, show that

$$1 + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4}\cdots = \frac{\pi^4}{90}\,.$$

[6]

SOLUTION

i) Sketch at end.

ii) The function is even, therefore $b_n = 0$. For a_n we can use the $\frac{1}{2}$-range series:

$$a_n = 2\int_0^1 (x-1)\cos(n\pi x)\,dx$$

$$a_0 = 2\left\{\left[(x-1)\frac{\sin(n\pi x)}{n\pi}\right]_0^1 - \int_0^1 \frac{\sin(n\pi x)}{n\pi}\,dx\right\}$$

$$= 2\left[(x-1)\frac{\sin(n\pi x)}{n\pi} + \frac{\cos(n\pi x)}{(n\pi)^2}\right]_0^1 = 2\left[0 - 0 + \frac{\cos(n\pi)}{(n\pi)^2} - \frac{1}{n\pi^2}\right]$$

$$= \begin{cases} -\frac{4}{(n\pi)^2}, & \text{n odd} \\ 0 & \text{n even} \end{cases}$$

$$\Longrightarrow f(x) = -\frac{1}{2} - \frac{4}{\pi^2}\sum_{\text{n odd}}^{\infty}\frac{1}{n^2}\cos(n\pi x)$$

$$= -\frac{1}{2} - \frac{4}{\pi^2}\sum_{m=1}^{\infty}\frac{1}{(2m+1)^2}\cos\left[(2m+1)\pi x\right]$$

Many errors resulting from integrating over $[-1,1]$ *instead of the half-range series, or people who, as in Q1(j), found the cancellation in* $\cos(n\pi) - 1$ *difficult to handle. Many of these then tried to force their result to resemble the given answer, without the correct derivation.*

iii) From Parseval's theorem

$$2\int_0^1 (x-1)^2\,dx = \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} a_n^2$$

$$\frac{2}{3} = \frac{1}{2}(-1)^2 + \sum_{\substack{n=1 \\ n\ odd}}^{\infty} \left(-\frac{4}{n\pi}\right)^2$$

$$\frac{1}{6} = \frac{16}{\pi^2}\sum_{\substack{n=1 \\ n\ odd}}^{\infty} \frac{1}{n^4} \Longrightarrow \frac{\pi^4}{96} = \sum_{\substack{n=1 \\ n\ odd}}^{\infty} \frac{1}{n^4}$$

We need

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \sum_{\substack{n=1 \\ n\ odd}}^{\infty} \frac{1}{n^4} + \sum_{\substack{n=1 \\ n\ even}}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{96} + \sum_{m=1}^{\infty} \frac{1}{(2m)^4} = \frac{\pi^4}{96} + \frac{1}{16}\sum_{n=1}^{\infty} \frac{1}{n^4}$$

Where we have reindexed. We have finally

$$\left(1 - \frac{1}{16}\right)\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{96} \Longrightarrow \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{16}{15}\left(\frac{\pi^4}{96}\right) = \frac{\pi^4}{90}$$

The most difficult question on the exam, with exactly three fully correct answers. Quite a lot of people got as far as the sum over odd n, but stopped there. For the l.h.s of the theorem, a lot of people integrated $(x-1)^2$ *over* $[-1,1]$ *which led to errors as the function is only* $x-1$ *on the interval* $[0,1]$*, and the even extension on* $[-1,0]$ *is* $-x-1$*!*

b) Obtain the nth derivative of the functions

$$(i)\ e^{-2x}, \qquad (ii)\ x\ln(x),$$

where $n > 1$. [Simplify the answer as much as possible.] [6]

SOLUTION

(i)

$$y = e^{-2x}$$
$$y' = -2e^{-2x}$$
$$y'' = 4e^{-2x}$$
$$\vdots$$
$$y^{(n)} = (-2)^n e^{-2x}$$

(ii) We need Leibnitz's formula:

$$y^{(n)}(x) = x\frac{d^n}{dx^n}(\ln x) + n(1)\frac{d^{n-1}}{dx^{n-1}}(\ln x) + 0 + 0\ldots$$

We have already seen $\ln(x+1)$ in Q3:

$$\rightarrow \qquad \frac{d^n}{dx^n}(\ln x) = (-1)^{n-1}\frac{(n-1)!}{x^n}$$

$$\rightarrow \qquad \frac{d^{(n-1)}}{dx^{(n-1)}}(\ln x) = (-1)^{n-2}\frac{(n-2)!}{x^{n-1}}$$

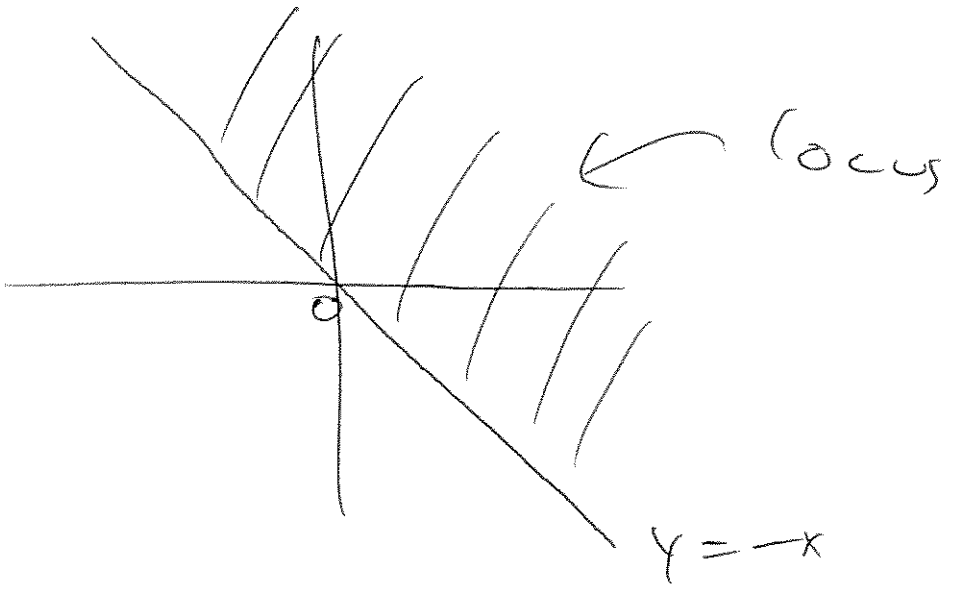
to give

$$y^{(n)}(x) = (-1)^{n-1}\frac{(n-1)!}{x^{n-1}} - (-1)^{n-1}\frac{n(n-2)!}{x^{n-1}} = (-1)^{n}\frac{(n-2)!}{x^{n-1}}$$

Where we have used $(n-1)! - n(n-2)! = (n-2)! \cdot [n-1-n]$, valid for $n \geq 2$.

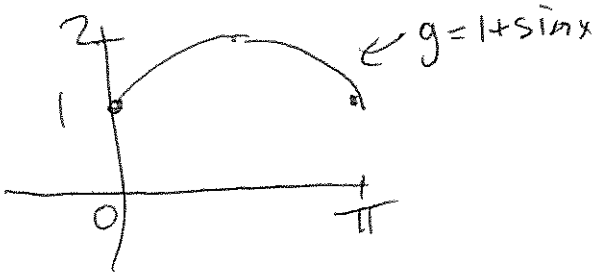
Most people found an easier approach for this one, just differentiating a few times and spotting the pattern! Unfortunately, a large number of those that tried to spot the pattern got it slightly wrong due to sloppiness. In two cases, the result was shown with a full 2-page proof by induction! Well done, but not required.

Q 1 (b)

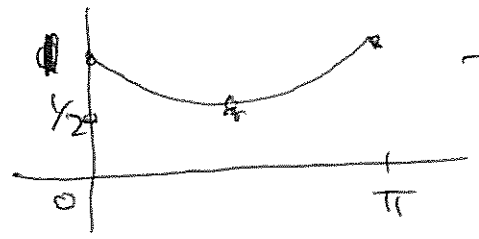


Q 3 (a)(ii)

Easiest:



$\Rightarrow$



$f = \frac{1}{1+\sin x}$

Q 4 (a)(i)

